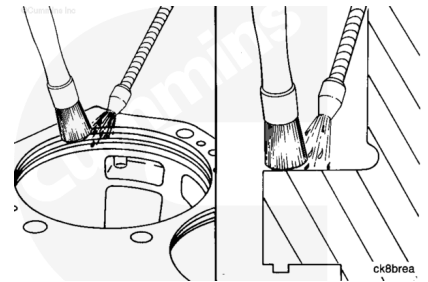


## Trainings ▾

## Install ▾

### **⚠ WARNING ⚠**

**When using solvents, acids, or alkaline material for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.**



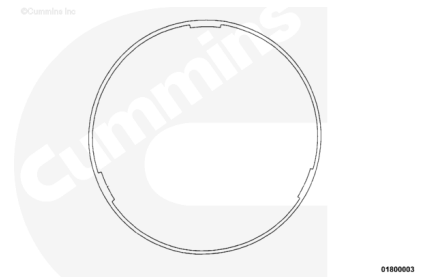
Clean the bottom of the cylinder liner flange with safety solvent.

**Note :** The seal rings have three locating tabs on the inside diameter. The tabs have an interference fit to the liner lower press fit diameter, to hold the seal ring in place during liner installation.

Install the seal rings.

The seal ring **must** be straight on the liner upon installation. Use your fingers to push the seal ring near the tabs to fit the seal ring down and over the lower press fit diameter during installation.

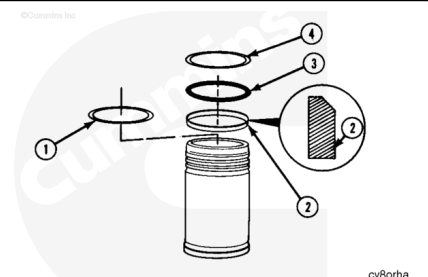
This practice during installation of the seal ring will prevent deformation that will result in the seal ring **not** fitting squarely on the bottom of the liner flange.



**Note :** Some o-rings have a "D" shape cross section. This type of o-ring **must** be installed with the flat side against the cylinder liner.

Install the crevice seal. The beveled edge of the crevice seal (2) **must** be positioned as shown.

Install o-rings in the position shown. Use the mold mark on the o-ring to check if the o-ring is twisted.



1. Liner, counterbore sealing ring
2. Crevice seal
3. Black o-ring

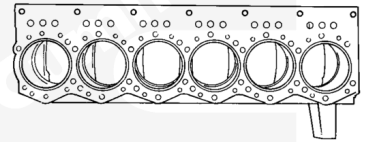
#### 4. Red o-ring.

### ⚠ CAUTION ⚠

**Do not use petroleum-based oil. This will cause swelling of the packing ring.**

Use vegetable oil to lubricate the inside diameter of the cylinder block packing ring bores.

Push the cylinder liners into the block with hand pressure.

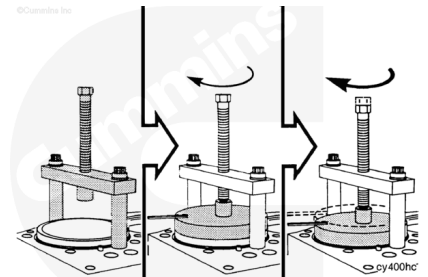


ck4breb

Use liner installation tool, Part Number, 3375422, or equivalent to install the bridge assembly and two cylinder head capscrews. Tighten the capscrews.

**Torque Value:** 65 N·m [48 ft·lb]

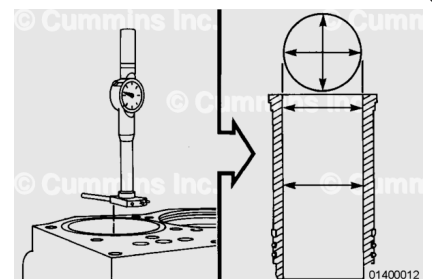
Install the pusher plate in the liner. Make sure it is aligned correctly in the liner. Turn the pusher screw until it touches the plate. Turn the pusher screw until the liner flange touches the counterbore ledge. Do **not** use more than 65 N·m [48 ft·lb] of torque. Remove the tool.



**Note :** New cylinder liners can be 0.005 mm to 0.015 mm [0.0002 in to 0.0006 in] smaller than the minimum specifications because of the Lubrite™ coating.

Use a dial bore gauge and measure the inside diameter of the liner at the top, bottom, and middle of the liner.

Perform two measurements at each location. The measurements **must** be 90 degrees apart.



| New Cylinder Liner Inside Diameter |     |        |
|------------------------------------|-----|--------|
| mm                                 |     | in     |
| 158.737                            | MIN | 6.2495 |
| 158.775                            | MAX | 6.2510 |

The inside diameter **must** be no more than 0.076 mm [0.003 in] out-of-round at the top two measurements.

If the inside diameter is more than 0.05 mm [0.002 in] out-of-round in the bottom measurement location, the liner **must** be removed. Check for a twisted o-ring.

Use the following procedure if machining is required. Refer to  
Procedure 001-058 in Section 1.  
(/qs3/pubsys2/xml/en/procedures/20/20-001-058.html)

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**Last Modified: 13-Aug-2015**

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